Cardiovascular System (CARD) · NPLEX-P1-CARD-14

CV-14 Cardiovascular Genetics

Work each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Space the two brain dumps for a competency across different days rather than doing them back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-14-01-C01				
NPLEX-P1-CARD-14-01-C02				
NPLEX-P1-CARD-14-01-C03				
NPLEX-P1-CARD-14-01-C04				
NPLEX-P1-CARD-14-01-C05				
NPLEX-P1-CARD-14-01-C06				
NPLEX-P1-CARD-14-01-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-14-01-C01-BD1

Draw the LDL receptor pathway and its feedback

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C01 Activity: NPLEX-P1-CARD-14-01-C01-BD1 Date: _____

Without looking at your notes, draw the path of one LDL particle from plasma to free cholesterol inside a hepatocyte. Label the apolipoprotein on LDL that the receptor binds, the coated pit, the acidified endosome, the recycling receptor, and the lysosome, and write what percentage of plasma LDL the liver clears. Beside the drawing, add the feedback limb: show where SCAP, SREBP-2, and INSIG sit when endoplasmic reticulum cholesterol is low and when it is high, and list the three genes the active SREBP-2 fragment switches on. Then write two sentences on what PCSK9 does to a receptor that has just delivered its cargo, and what that means for how many trips each receptor makes. Finish by naming the other apolipoprotein the receptor binds and the lipoprotein that carries it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-14-01-C01-BD2

Why plasma LDL stays high while the liver cell starves

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C01 Activity: NPLEX-P1-CARD-14-01-C01-BD2 Date: _____

A man has untreated LDL cholesterol of 310 mg/dL (8.0 mmol/L) with normal triglycerides, tendon xanthomas at the Achilles, and corneal arcus at age 40. His father had a myocardial infarction at 45. Write the inheritance pattern and the three genes that can produce this picture, and for each gene write whether the variant is loss of function or gain of function. Then explain in four or five sentences why his hepatocytes behave as though they are short of cholesterol while his plasma is overloaded, and why that keeps new cholesterol synthesis and VLDL secretion going. Next, write what happens to his plasma LDL if his cells are given cholesterol by a route that does not use the receptor, and what that result proves about which step is broken. Then write why lowering hepatic synthesis helps him more than it would help a child with no working receptors at all. Finish by writing which receptor on macrophages keeps taking up oxidized LDL no matter how much cholesterol the cell already holds, and what that produces in skin, tendon, and artery wall.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-14-01-C01-RT01

Rebuild the familial hypercholesterolemia sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C01 Activity: NPLEX-P1-CARD-14-01-C01-RT01 Date: _____

Rebuild the six-step sequence that runs from a variant in one of the three genes to xanthomas and premature atherosclerosis, one step per line and in order. Beside the step about receptor trips, write why a shortage of receptors raises LDL production as well as lowering LDL clearance. Beside the step about the endoplasmic reticulum, write which transcription factor stays active and name the rate-limiting enzyme it keeps switching on. Then add a line under the sequence naming the five classes of LDL receptor variant by the step in the receptor's life that fails, and mark which class is the most common.

Rebuild from memory: Pathogenesis of familial hypercholesterolemia

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-14-01-C02-BD1

Sarcomere genes to the hypertrophic cardiomyopathy phenotype

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C02 Activity: NPLEX-P1-CARD-14-01-C02-BD1 Date: _____

Write the definition of hypertrophic cardiomyopathy, its approximate prevalence in adults, and its inheritance pattern. Name the two genes that account for most genotype-positive cases, write the protein each one encodes, and write which one acts through a dominant negative mechanism and which through haploinsufficiency, with the type of variant behind each. List the less common sarcomere genes from the notes. Then write the share of patients in whom testing finds a variant, and one sentence on what a negative test does and does not mean. Next, rebuild in six steps how a faulty sarcomere protein leads to a stiff, nondilated, arrhythmia-prone ventricle, and beside the contractility step write the two changes at the level of the myosin head and the thin filament that make the heart hypercontractile and wasteful of ATP. Finish with the morphology: wall thickness that is diagnostic against the normal value, the pattern of hypertrophy and cavity size, the hallmark microscopic finding, and the two other tissue changes that accompany it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-14-01-C02-BD2

Two relatives, one variant, and a murmur that moves

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C02 Activity: NPLEX-P1-CARD-14-01-C02-BD2 Date: _____

A woman with a pathogenic MYH7 variant has a septal wall thickness of 19 mm and exertional dyspnea. Her 16-year-old son carries the same variant and has a normal echocardiogram. Write the recurrence risk for each of her children, then explain in three or four sentences why the son can carry the variant and look normal, naming the two genetic terms involved and saying what makes penetrance age dependent. Write what you would do with his evaluation over time and why, and write which gene is most likely to delay hypertrophy into middle age. Next, write why the mother's symptoms and her fourth heart sound follow from the way the ventricle fills, and why atrial fibrillation would make her much worse. Then take her systolic murmur: write what happens to it on standing and during the strain of a Valsalva maneuver, what happens on squatting, leg raising, and handgrip, and give the mechanism in terms of ventricular volume, the septum, and the anterior mitral leaflet. Write how the murmur of aortic stenosis responds to the same maneuvers. Finish by writing why she can have angina with normal coronary arteries, and by listing the conditions from the notes that can imitate this picture with a line each on what separates them.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-14-01-C02-CA01

Collapse during a basketball game in a 17-year-old

A 17-year-old boy collapses during a basketball game and recovers consciousness within a minute. He has had exertional breathlessness for two years that he attributed to being out of condition. A paternal uncle died suddenly at age 29. Temperature is 37.1 C (98.8 F), pulse is 76/min and regular, respirations are 16/min, and blood pressure is 118/72 mm Hg. A fourth heart sound is present. A systolic murmur is heard at the left sternal border; it becomes louder when he stands up from a squat and softer when he squats again or grips a handgrip dynamometer. The electrocardiogram shows high voltage with deep narrow Q waves in the lateral leads. Echocardiography shows asymmetric septal thickening of 20 mm, a small left ventricular cavity, systolic anterior motion of the mitral valve with mitral regurgitation, and normal left ventricular emptying. He trains year round.

1. Recognize. Name the condition, its inheritance pattern, and the two genes that most often cause it, writing the protein each one encodes.

2. Explain. Explain how a variant in one of those proteins produces a thick, nondilated ventricle when loading conditions are normal, taking the mechanism from the protein through contractility and energy use to hypertrophy and fibrosis.

3. Differentiate. Write how this picture differs from athlete's heart, from hypertensive hypertrophy, and from the other phenocopies in the notes, naming the finding that separates each one.

4. Investigate. Write which features of the murmur, the electrocardiogram, and the tissue you would examine to support this diagnosis, and what value or appearance each would have.

5. Decide. State which tissue changes in this ventricle create the substrate for the arrhythmia that caused his collapse, and state what determines the risk to his first-degree relatives.

6. Justify. Justify the behavior of his murmur from ventricular volume, the position of the septum, and the anterior mitral leaflet, and explain why his uncle's variant could be present in relatives who have a normal echocardiogram.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-14-01-C03-BD1

Marfan syndrome from gene to dissection

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C03 Activity: NPLEX-P1-CARD-14-01-C03-BD1 Date: _____

Write what the aortic media is built from, naming the elastic lamellae, the cells between them, and the two matrix components that hold them. Then write the gene, chromosome, and protein in Marfan syndrome, its prevalence, the proportion of cases that are new variants, and the inheritance pattern. Write the two normal jobs of that protein, including what it anchors in the eye and the growth factor it holds in a latent form. Rebuild in six steps the sequence from the variant to aortic regurgitation and dissection, and beside the dilation step write Laplace's law in words and explain why dilation feeds further dilation. Name the lesion seen in the wall, list its three microscopic features, and write the older term for it and why that term is misleading. Finish by listing the features outside the vessels that come from pleiotropy, including the direction the lens is displaced, and write how homocystinuria differs on three points.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-14-01-C03-BD2

Three families, three failing wall components

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C03 Activity: NPLEX-P1-CARD-14-01-C03-BD2 Date: _____

Three unrelated patients each have arterial disease. Patient 1 has a dilated aortic root at the sinuses of Valsalva, an upward displaced lens, and long fingers. Patient 2 at age 22 has aneurysms at several sites with tortuous arteries, widely spaced eyes, and a split uvula, and his aorta dissected at a diameter smaller than usually expected. Patient 3 has thin translucent skin, easy bruising, and a dissection of a medium-sized artery with no preceding aneurysm, and a sister died of sigmoid colon rupture. For each patient write the gene, the protein or receptor involved, and the inheritance pattern, then explain in two or three sentences how the failing wall component produces that particular arterial pattern. For patient 2, write the paradox between what the variant does to receptor function in cells and what happens to signaling in the aortic wall. For patient 3, write which amino acid in the Gly-X-Y repeat is usually replaced, why no other amino acid fits that position, and what happens to the triple helix. Then work the arithmetic: if mutant and normal chains are made in equal amounts and assemble at random, calculate the fraction of fully normal molecules for a three-chain collagen and for a four-subunit potassium channel, and compare each with the fraction left by a null allele. Use those numbers to explain why a missense variant is often more severe than a complete loss of one allele.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-14-01-C03-CP01

The three heritable disorders of the arterial wall

Fill in every cell of the table from memory, then open the notes and correct anything you missed.

Feature	Marfan syndrome	Loeys-Dietz syndrome	Vascular Ehlers-Danlos syndrome
Gene or genes and the protein affected			
Inheritance pattern			
Where in the arterial tree the disease appears, and at what age and diameter			
Whether an aneurysm precedes rupture or dissection			
What goes wrong in the wall at tissue level			
Findings in the eye			
Findings in the face, mouth, and skull			
Findings in skin, bowel, and uterus			
Skeletal and joint findings			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-14-01-C04-BD1

Repolarizing currents and the long QT genotypes

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C04 Activity: NPLEX-P1-CARD-14-01-C04-BD1 Date: _____

Draw the ventricular action potential with voltage on the vertical axis and time on the horizontal axis, and label phases 0 to 4. Write beside phase 3 the two currents that do most of the work there, giving the gene for each and the accessory subunit that partners one of them, and name the current that finishes repolarization. Explain in three or four sentences what repolarization reserve means and why a carrier can have a resting QT interval that is barely long, then list the four situations from the notes that can unmask the problem. Next, build the table of the three main long QT genotypes from memory with one row each, using the columns gene, direction of the functional change, the current affected, the typical trigger, and the appearance of the T wave. Mark which genotype is the most common. Finish by writing the one sentence that captures the pattern across the three genes, separating what happens to the potassium channel genes from what happens to the sodium channel gene.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-14-01-C04-BD2

Why the trigger, the ear, and the fever all follow from the current

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C04 Activity: NPLEX-P1-CARD-14-01-C04-BD2 Date: _____

Three young people have events. One faints while swimming, one collapses when a fire alarm sounds, and one is found dead in bed. For each, write the likely genotype and explain from the behavior of the affected current why the event happened in that setting rather than another. Then write what deranges rhythm at the cellular level in all three, naming the afterdepolarization and the arrhythmia it produces. Next, a deaf child has a QT interval above 500 ms and two affected alleles of the same gene. Write the syndrome name and its inheritance pattern, name the common dominant form with normal hearing, and explain why the same channel that repolarizes the ventricle is also needed for hearing, naming the structure in the ear and the ion it secretes. Write why both parents hear normally and what their electrocardiograms may still show. Finish with one gene that produces two different diseases: write the gene, which functional change causes which disease, and what the term for this is.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-14-01-C04-RT01

Rebuild the Brugada sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C04 Activity: NPLEX-P1-CARD-14-01-C04-RT01 Date: _____

Rebuild the six-step sequence that runs from a loss of sodium current to ventricular fibrillation in Brugada syndrome, one step per line and in order. Beside the first step name the gene, the inheritance pattern, and the approximate share of patients in whom the variant is found. Beside the notch step name the current that deepens it and write why it matters that this current is large in right ventricular outflow epicardium and smaller in endocardium. Under the sequence, draw two action potentials on one set of axes, one epicardial and one endocardial, and show the lost dome; then draw the surface tracing in V1 and V2 and label the shape of the ST segment, the minimum elevation that counts, and the direction of the T wave. Finish by listing the three things from the notes that can unmask the pattern, with one line on why warming does it, and write the two features of the patient population in which this condition is most common.

Rebuild from memory: Brugada syndrome from SCN5A loss to ventricular fibrillation

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-14-01-C04-PR01

Predict the effect of a low plasma potassium in LQT2

A woman with a KCNH2 variant and a baseline QT interval at the upper limit has been vomiting for three days. Her plasma K+ is 2.7 mEq/L (reference range 3.5 to 5.0 mEq/L). Predict each item below, write the direction of the change, and give the current, channel, or mechanism that explains it in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the rapid delayed rectifier current IKr carried by her remaining normal channels		
What happens to the margin left by repolarization reserve		
What happens to the duration of the ventricular action potential and to the QT interval		
What happens to L-type Ca2+ channels late in the prolonged plateau		
What kind of afterdepolarization appears, and at which phase of the action potential		
What the resulting rhythm is called and what it can degenerate into		
What happens to her risk if she is also given a drug that blocks IKr		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-14-01-C05-BD1

Embryology to the characteristic heart defect

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C05 Activity: NPLEX-P1-CARD-14-01-C05-BD1 Date: _____

Write what the endocardial cushions do during development, naming the canal they divide, the opening they close, and the valves they help build. Then write what the neural crest cells contribute and which vessel their septum divides. Next, take each of these in turn: Down syndrome, Turner syndrome, the 22q11.2 deletion, Williams syndrome, and Noonan syndrome. For each one write the genetic basis, the characteristic cardiovascular lesion, and the embryonic process or protein that links the two, then add the findings outside the heart from the notes. Include for Down syndrome the three mechanisms by which the trisomy arises with their approximate shares and which parental meiotic division is usually involved, and the proportion of patients who have a heart defect. Include for Turner syndrome the karyotypes, the frequency of each left-sided lesion, and the hypothesis about fetal lymphedema. Include for the 22q11.2 deletion the size of the deletion, the gene whose single copy matters, and the two pouch derivatives that fail with the consequence of each. Include for Williams syndrome the gene and what elastin tells smooth muscle to do.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-14-01-C05-BD2

Read the findings back to the mechanism

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C05 Activity: NPLEX-P1-CARD-14-01-C05-BD2 Date: _____

Five infants and children present. Child 1 has hypotonia, a flat face, duodenal atresia, and a single murmur with a common atrioventricular valve on echocardiography. Child 2 has hypocalcemic seizures, recurrent infections, a cleft palate, and an interrupted aortic arch. Child 3 is a girl with short stature, a webbed neck, blood pressure higher in the arms than the legs, and weak delayed femoral pulses. Child 4 has elfin facies, a high calcium level, and a systolic murmur with narrowing just above the aortic valve. Child 5 has a normal karyotype, features that resemble Turner syndrome, a dysplastic stenotic pulmonary valve, and a thick ventricle. For each child name the condition, the genetic basis, and the embryonic or cellular mechanism that produced the cardiac lesion. For child 1, write why a large shunt in this syndrome reaches pulmonary hypertension sooner than the same shunt in another child. For child 2, write why the immune and parathyroid findings travel with the cardiac findings. For child 3, write the three physical and radiographic consequences of her aortic lesion and what her aorta is at risk of later. For child 4, write why a variant in that one gene alone gives a narrower picture than the deletion does. Then write the difference between a syndrome, an association, and a sequence, give the example of each from the notes, state how many features are needed for the association and what its recurrence risk is, and name the two single-gene conditions that imitate it with the feature that gives each one away.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-14-01-C05-CP01

Genetic syndromes and their cardiovascular lesions

Fill in every cell of the table from memory, then open the notes and correct anything you missed.

Syndrome	Genetic basis	Characteristic cardiovascular lesion	Developmental process disturbed	Other findings that point to it
Down syndrome				
Turner syndrome				
22q11.2 deletion syndrome				
Williams syndrome				
VACTERL				
Noonan syndrome				

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-14-01-C05-Q01

A newborn with hypocalcemia and an arch anomaly

A newborn boy has jitteriness and a seizure on the second day of life. Serum calcium is low. Examination shows a cleft palate and absent thymic tissue on imaging. Echocardiography shows interruption of the aortic arch distal to the left subclavian artery. Which of the following best explains the combination of findings in this infant?

A. Trisomy of chromosome 21 causing failure of endocardial cushion fusion

B. Haploinsufficiency of elastin causing smooth muscle proliferation in the arterial media

C. Loss of one copy of TBX1 disturbing the pharyngeal arches and the neural crest

D. An expansion of a trinucleotide repeat silencing a mitochondrial iron handling protein

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-14-01-C06-BD1

Dystrophin, frataxin, and mitochondrial DNA

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C06 Activity: NPLEX-P1-CARD-14-01-C06-BD1 Date: _____

Draw the link from actin inside the myocyte to laminin in the basement membrane, labelling dystrophin and the two dystroglycans, and write what that link protects during contraction. Write the gene, its chromosome, why so many cases are new variants, and the inheritance pattern, then give the prevalence of Duchenne dystrophy in males and the proportion of variants that are deletions. State the reading frame rule in full: what an in-frame deletion produces and which disease it gives, what a frameshift produces and which disease it gives, and how often the rule holds. Then write the pathogenesis from a torn membrane to myocyte necrosis, the laboratory finding, the two physical signs from the notes, and the cardiac picture including where fibrosis starts, what cardiomyopathy results and by what age, and the three electrocardiographic findings. Next, write the gene, the repeat, its position in the gene, the normal and disease repeat ranges, and the inheritance pattern for Friedreich ataxia, then write what the missing protein is needed for and which complexes and enzyme depend on it, which tissues are hit and why, and what the cardiomyopathy looks like. Finish with mitochondrial DNA: its approximate size, what it encodes, how it is inherited, and the three mechanisms that make the mutant fraction differ between tissues and siblings, then write the syndrome from the notes with its genetic lesion, its age of onset, its three defining features, and its conduction consequence with the muscle biopsy finding.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-14-01-C06-BD2

Work the frame, the carrier, and the threshold

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C06 Activity: NPLEX-P1-CARD-14-01-C06-BD2 Date: _____

Two boys have DMD deletions. Boy A has lost exons totaling 522 coding nucleotides and boy B has lost 197. For each boy, divide and show your working, state whether the downstream reading frame survives, state what protein is made, and name the disease, then write which boy has the larger deletion and why that does not predict the milder disease. Write what happens to the transcript when a premature stop codon appears. Next, a 32-year-old woman whose son has Duchenne dystrophy has no limb weakness but has a dilated, poorly contracting left ventricle. Explain in three or four sentences how a carrier of an X-linked recessive disease develops cardiomyopathy, naming the process in her myocytes and what happens to the cells that express only the mutant allele. Write why her heart cannot replace them. Then two sisters carry the same mitochondrial DNA variant and only one has cardiac disease. Explain the result using heteroplasmy, random segregation, the oocyte bottleneck, and tissue threshold, and write which tissues have the lowest threshold and why. Finish by writing why membrane fragility tends to give a dilated ventricle, why a failure of energy supply tends to give a thick one, and why a large mitochondrial DNA deletion shows itself first in the conduction system.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-14-01-C06-Q02

Progressive block in a young adult

A 19-year-old woman has had drooping eyelids and limited eye movements since childhood. Funduscopy shows pigmentary changes in the retina. The electrocardiogram shows a widening QRS complex with progressive atrioventricular block, and muscle biopsy shows ragged red fibers on modified Gomori trichrome staining. Her mother and two brothers are unaffected. Which of the following best accounts for her findings?

A. A large deletion of mitochondrial DNA removing respiratory chain subunits and transfer RNAs

B. A frameshift deletion in DMD on the X chromosome abolishing dystrophin

C. A GAA repeat expansion in the first intron of FXN silencing transcription

D. A missense variant in MYH7 acting as a dominant negative in the thick filament

Why did you pick it?

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-14-01-C07-BD1

Why the adult heart grows by hypertrophy

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C07 Activity: NPLEX-P1-CARD-14-01-C07-BD1 Date: _____

Write what happens to cardiomyocyte division around birth and what many of the cells become instead, then list the four molecular brakes that hold the cells out of the cell cycle. Write the measured renewal rate in young adults, the method that produced that figure, and the term for a tissue that heals by scar, then contrast cardiac muscle with skeletal muscle and smooth muscle on their capacity to replace lost cells. Next write Laplace's law for ventricular wall stress as a relationship between pressure, radius, and wall thickness, then write for pressure overload and for volume overload the load that causes it, where the new sarcomeres are added, the shape of hypertrophy that results, and what each one does to wall stress. Give normal heart weight for women and for men. Then write how stretch is sensed, naming three sensors, and trace the pathologic signaling route from angiotensin II, endothelin-1, and norepinephrine through the receptor family, the enzyme, the second messengers, the phosphatase, and the transcription factors that reach the nucleus. Write the separate route used by exercise hypertrophy. Finish with the fetal gene program: name the genes switched on and the two switched down, and write the functional consequence of each of those changes, including what the natriuretic peptides do and the value of BNP below which heart failure is an unlikely cause of acute breathlessness.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-14-01-C07-BD2

Two thick hearts and one scar

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C07 Activity: NPLEX-P1-CARD-14-01-C07-BD2 Date: _____

A 28-year-old rower and a 58-year-old man with untreated hypertension both have increased left ventricular wall thickness. For each, write the signaling route that produced it, whether the fetal gene program is switched on, what the capillary supply and the interstitium look like, and what happens to the hypertrophy if the stimulus is removed. Then explain in three or four sentences why one of them is at risk of progressing to heart failure and the other is not, and why the ventricle that keeps losing myocytes eventually dilates even though it started by thickening. Next, take a man six weeks after a myocardial infarction. Write the sequence of repair from macrophage clearance through the growth factor released, the cell type that appears, the first collagen laid down, and the collagen that replaces it. Write the two things the finished scar cannot do and the two consequences of each for pump function and for rhythm. Write which forms of injured myocardium can recover and which cannot. Finish by writing why every disease in this chapter that kills myocytes ends in the same three findings, and name two diseases from earlier in the chapter where that repair follows myocyte loss.

My response

Confidence before checking

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Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-14-01-C07-RT01

Rebuild the maladaptive hypertrophy cycle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-14 Cardiovascular Genetics Competency: NPLEX-P1-CARD-14-01-C07 Activity: NPLEX-P1-CARD-14-01-C07-RT01 Date: _____

Draw the six-step cycle of pathologic hypertrophy as a closed loop, numbering each step and writing it out in words, starting from a sustained load that raises wall stress. Beside the step on gene expression, name the genes switched on. Beside the step on the interstitium and the calcium pump, write what each change does to relaxation and to oxygen supply. Beside the step on myocyte death, write why the survivors have more work to do. Beside the closing step, write Laplace's law in words and explain why dilation sends the loop round again. Underneath the loop, write the one sentence about cardiomyocyte renewal that explains why the loop cannot be broken by replacing the lost cells.

Rebuild from memory: Maladaptive cycle of pathologic hypertrophy

My response

Confidence before checking

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